



PROJECT CHARTER

Group Members:

Jaime F. De la Pena

Danielle Heart L. Gonzales

Russell John Reinaldi M. Salvador

Gabriel Quenneth T. Vasquez

1. Project Title

- **Postfolio:** Schema-Driven Internal Tool Builder

2. Project Purpose / Justification

Small and medium enterprises (SMEs) require internal tools that adapt to their specific operational workflows. Most internal tools are either too rigid (SaaS) or too ugly (custom-built). Postfolio provides a premium, modular platform allowing businesses to build everything from POS terminals and inventory trackers to staff dashboards without writing a single line of frontend code, utilizing a schema-first and UI-driven approach.

3. Project Description / Scope

Inclusions:

- **Database Editor:** Aupabase-inspired interface for defining domain models and database schemas directly within the platform.
- **UI Builder:** A professional, Figma-like visual design environment for creating staff-facing screens using modular drag-and-drop elements.
- **Terminals (In-points):** Isolated, secure deployments of UI layouts authenticated via robust PIN access.
- **Hybrid Database Architecture:** Utilizing Supabase for a centralized database.

Exclusions:

- Native mobile applications (iOS/Android).



- Complex third-party payment gateway integrations.

4. Objectives

- Develop a schema-first database editor that abstracts DDL operations.
- Build a functional, highly responsive drag-and-drop UI builder capable of rendering user-specific layouts.
- Implement a secure, PIN-based terminal access system with robust client and server-side validation.
- Achieve seamless toggling between local SQLite persistence and remote Supabase storage without altering frontend logic.

5. High-Level Requirements

- **Functional:** The system must allow users to define data models. The UI Builder must map directly to database schemas. The Terminals must lock behind a 4-8 digit PIN and render layouts reactively. The application must feature a developer mode toggle for backend storage.
- **Technical:** Built with the Bun runtime, Nuxt 3 (Vue 3, Composition API) framework, and styled with Vanilla CSS and Tailwind CSS. The database layer utilizes SQLite locally and Supabase in production.

6. Assumptions and Constraints

- **Assumptions:** Users have basic knowledge of their business data requirements. The deployment environment supports modern web standards.
- **Constraints:** Time constraints of the academic semester. Zero budget for premium APIs or expensive cloud hosting outside of Supabase's free tier.

7. Major Deliverables

- System Architecture and Database ERD.
- UI/UX Wireframes and Prototype.
- Fully functional Postfolio software system with LAN synchronization and real-time event broadcasting.
- Project Final Report and User Manual.

8. Milestone Schedule

- Week 1-2: Finalize requirements, project charter approval, and database architecture design (multi-tenancy schema, per-business Postgres schemas, platform-level tables).



- **Week 3-4:** Set up Nuxt 3 + Bun project structure, Supabase integration, dev mode SQLite abstraction layer, and Supabase Auth for admin and in-point accounts.
- **Week 5-7:** Build the admin dashboard, table editor, schema visualizer, and Schema API routes (create, alter, drop tables per business schema).
- **Week 8-9:** Implement the free form UI canvas builder (drag-and-drop, snapping, element properties panel) and in-point system (role presets, PIN login, canvas runtime renderer).
- **Week 10:** End-to-end QA testing, schema API, auth flows, UI builder save/load, in-point rendering, and dev mode vs production mode parity.
- **Week 11:** Final project presentation.
- **Week 12 onwards:** Final revisions and sign-off.

9. Budget Estimate

- ₱0.00 (Development utilizes existing local hardware, open-source frameworks, and standard academic software licenses).

10. Project Team / Roles and Responsibilities

- Key members and their responsibilities.
 - **Project Manager:** Jaime F. De la Pena - Oversees schedule, tasks, and documentation.
 - **Lead Developer (Backend/DB):** Gabriel Quenneth T. Vasquez - Handles hybrid database architecture, backend endpoints, Nuxt server routes, and SQLite/Supabase integration.
 - **Lead Developer (Frontend/UI):** Danielle Heart L. Gonzalez - Implements the Figma-like UI builder, terminal interfaces, and PIN authentication UI.
 - **Quality Assurance / Tester:** Russell John Reinaldi M. Salvador - Conducts testing, verifies UI element rendering, terminal security, and ensures aesthetic consistency.

11. Stakeholders

- **Project Team:** Jaime F. De la Peña, Danielle Heart L. Gonzales, Russell John Reinaldi M. Salvador, Gabriel Quenneth T. Vasquez.
- **Course Instructor:** Elmerito D. Pineda
- **Target End-Users (SMEs):** Businesses needing modular POS and internal tools. (ex. Truffino)



12. Approval Requirements

- The system must pass a live demonstration showing database schema creation, UI building, and secure terminal login.
- Sign-off from the course instructor based on the project rubric.

13. Risks and Issues

- **Risk:** Synchronization issues between the UI Builder's canvas and internal component state.
- **Risk:** Ensuring seamless transitions between local SQLite and remote Supabase without breaking API routes.
- **Risk:** Maintaining high performance and responsiveness within the drag-and-drop canvas interface.

14. Signatures / Approval

Jaime F. De la Peña
Project Manager
Postfolio

Gabriel Quenneth T. Vasquez
Lead Developer
Postfolio

Russell John Reinaldi M. Salvador
Quality Assurance
Postfolio

Danielle Heart L. Gonzales
Lead Developer
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